

Surplus-to- *Shelter*

Real-time algorithmic food rescue engine.

Solving the 2–6 hour spoilage bottleneck with live capacity-matching & automated ESG receipts.

TEAM

TechCypher

DEPLOYMENT

Live Netlify Web App

URGENT

2–6

HOURS

SPOILAGE WINDOW

LATE

SURPLUS

SHELTER

Why good cooked food *still* becomes waste.

The missing ingredient is not supply, capacity, or goodwill. It is a clock that beats faster than WhatsApp, a gate we cannot see, and a margin that points the wrong way.

01 / 03

2–6

H

The spoilage window.

Cooked food decays faster than manual coordination can run. WhatsApp forwards lose the thread. Calls die on hold. By the time a driver would arrive, the tawa is cold.

02 / 03

0

/n

The capacity blind spot.

Shelters have free cold-storage slots we never see. A 30kg delivery to a shelter with 8kg of room is a re-dump an hour later, at the next chowk.

03 / 03

₹0

/kg

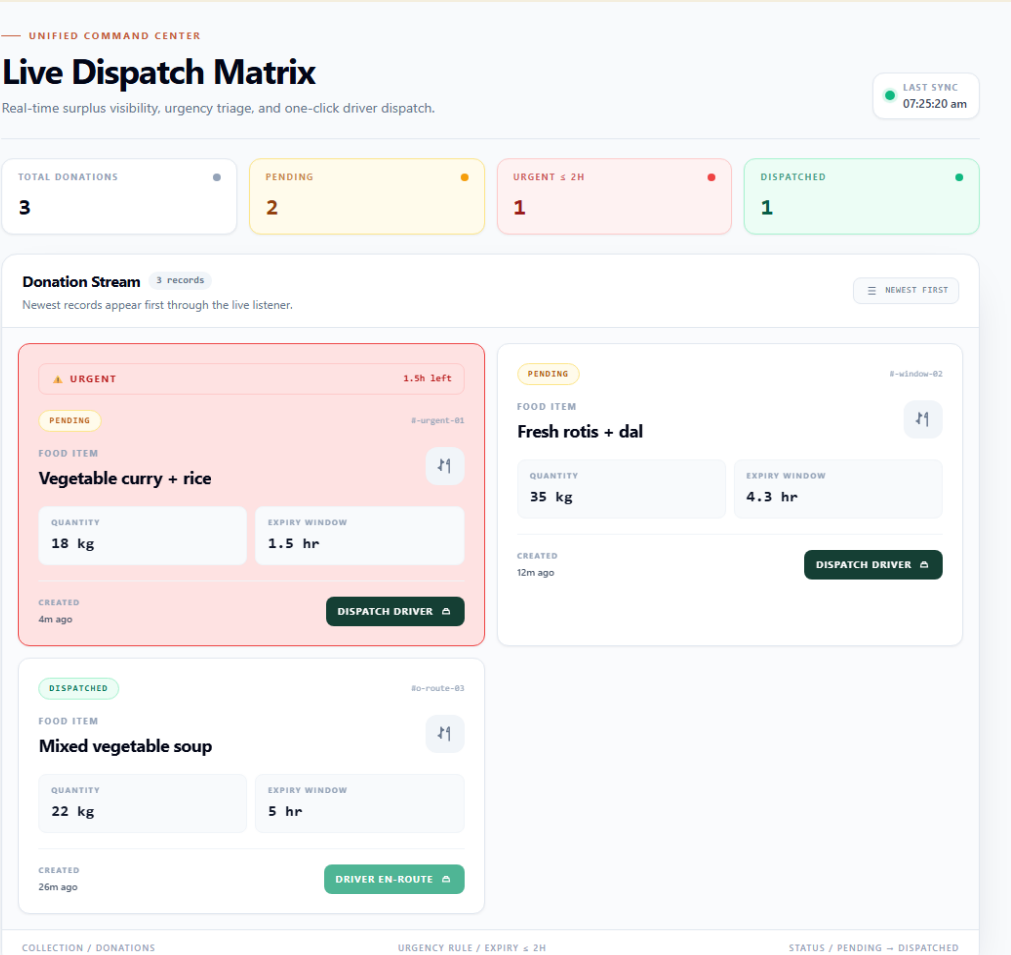
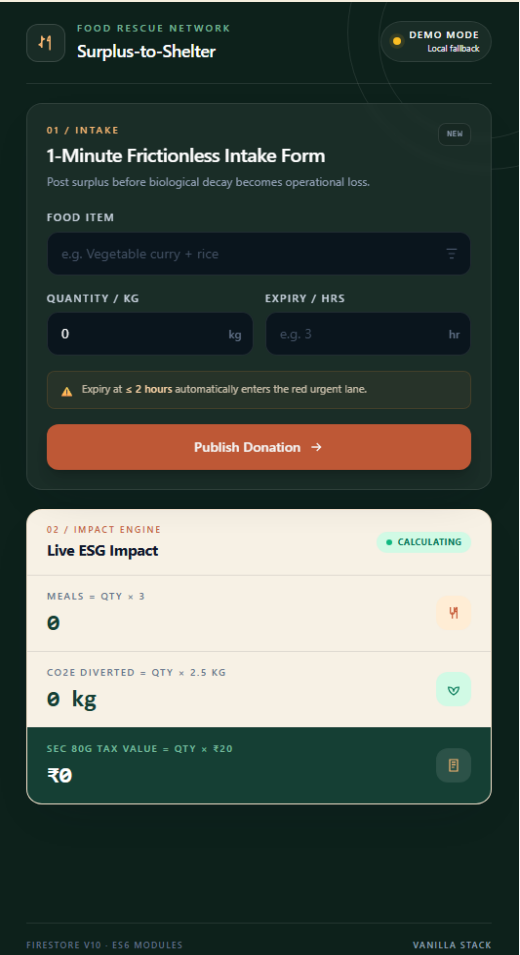
Cost goes the wrong way.

The dustbin is one tap away. Donating is twenty-five. No automated receipt, no 80G paper trail, no finance committee recognition. The trolley rolls past the donation shelf.

The gap is not distance. It is a missing 30-second data bridge.

Context / UNEP Food Waste Index 2024: 1.05B tonnes wasted globally in 2022 • 132 kg per capita • ~1/5 of food available to consumers.

The web app turns a rescue into a live dispatch decision.



MVP FLOW

01 / INTAKE **< 3 MIN**

Food item, kg, expiry window

02 / TRIAGE **< 2 HRS**

Urgent food flags in red

03 / DISPATCH **50 KG CAP**

Capacity-aware route; status flips to DISPATCHED

LIVE / DATA **LIVE SYNC**

onSnapshot real-time sync without page reloads

A pipeline that *beats biological decay.*

The product turns a donor handoff into a timed, capacity-aware decision.

DONOR INTAKE

EXPIRY TRIAGE

CAPACITY MATCH

DRIVER DISPATCH

01 / FEATURE

< 3 MIN

Frictionless donor intake form.

A 1-minute intake captures food item, quantity, and expiry window.

02 / FEATURE

< 2 HRS

Expiry triage engine flags urgent food in red.

The 2–6 hour spoilage window becomes a red, time-stamped decision lane.

03 / FEATURE

50 KG CAP

Capacity-aware dispatch prevents secondary dumping.

The cap keeps one shelter from receiving a bag it cannot safely store.

FAILURE HANDLING

Spoilage Cut-off

Auto-cancel dispatch if shelf-life drops under 45 mins. Zero liability.

Capacity Triage

Dynamic routing cascades to the next NGO if the 50kg limit is hit.

0-Lag Architecture

Firebase real-time sync without page reloads.

A live *Sec 80G* certificate, printed on every rescue.

Every kilogram that flows through the system produces an auditable handoff — and three real numbers the finance team can write into the next CSR return.

THE COMPUTATION / PER KILOGRAM RESCUED

Meals = Qty × 3

CO₂e = Qty × 2.5 kg

Tax value = Qty × ₹20

WORKED EXAMPLE / 10 KG RESCUE

30 meals

25 kg CO₂e diverted

₹200 impact value

IMMUTABLE AUDIT TRAIL

01 Posted ——— 02 Matched ——— 03 Dispatched

Each handoff is time-stamped and linked. The 80G certificate is generated from the same audit trail.

80G NOTE / ₹20 per kg is a configurable impact-value proxy. Actual deduction depends on the qualifying donation and recipient status.

Tax reference / Income Tax Department, Section 80G deduction guidance. Product note / the ₹20/kg value is a declared impact proxy, not a statutory tax rate.

From a Jaipur pilot to a

repeatable rescue network.

The product earns its place when both sides of the handoff get something measurable.

RESTAURANTS / KITCHENS

Donating becomes a 20-five tap process, not a twenty-five call.

Auto-issued ESG receipts give finance teams a clean CSR attachment and a reason to repeat the behavior.

SHELTERS / NGOS

Predictable inflow beats dropped bags at odd hours.

Menu planning becomes predictable; every delivery is time-stamped and linked to a safe handoff.

THE FLYWHEEL

SURPLUS

AUDIT

LIVE DATA

MEAL

MATCH

30-DAY PILOT TARGETS

≤ 3 MIN / intake

< 2 HRS / urgent response

0 / dropped bags

100% / logged handoffs

Pilot success = a measurable reduction in spoilage, a reliable NGO handoff, and a CSR-ready record for every kilogram.